Project Title:

Hotel Booking System(OOPs)

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Problem Statement and Target User

MANUAL BOOKING IS INEFFICIENT AND LEADS TO ERRORS LIKE DOUBLE BOOKING.

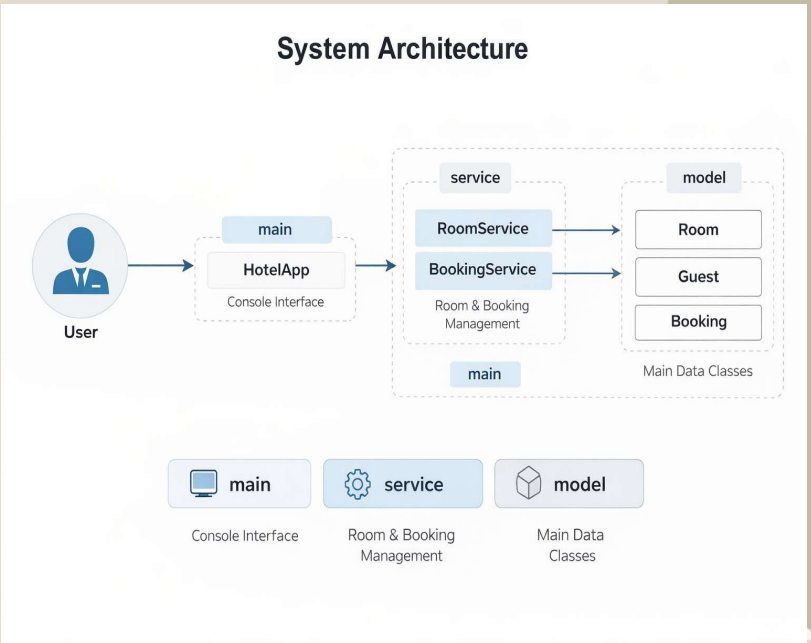
HOTEL STAFF

RECEPTIONIST

SYSTEM DESIGN

High-Level Architecture

- User → HotelApp → Services → Model



Key Classes:

Room (base class)

Single Room, Double Room, Suite Room

Guest , Booking

Room Service , Booking Service

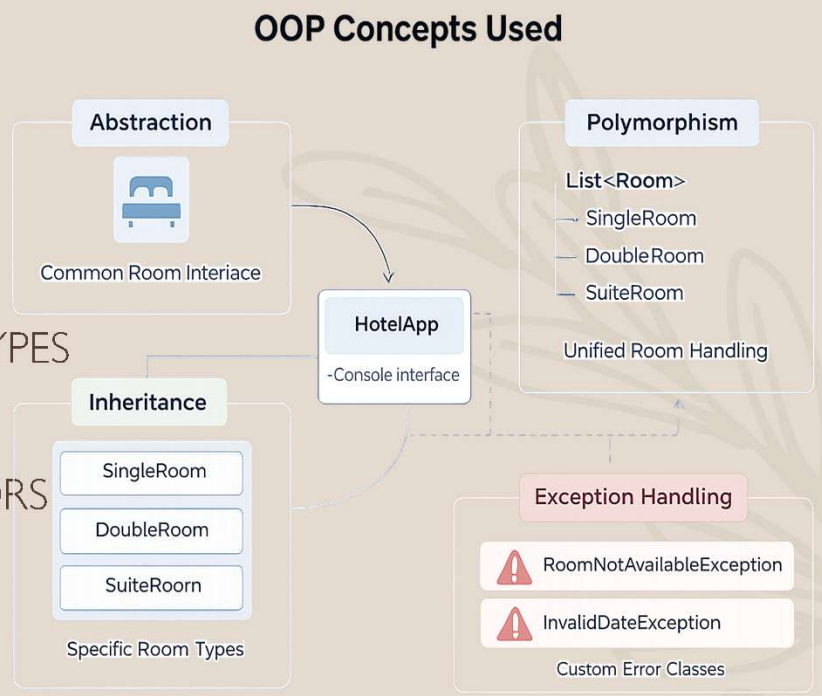
OOPS CONCEPTS

ABSTRACTION: ROOM CLASS

INHERITANCE : ROOM → SINGLE/DOUBLE/SUITE

POLYMORPHISM : ROOM REFERENCE → DIFFERENT ROOM TYPES

EXCEPTION HANDLING: CUSTOM EXCEPTIONS FOR ERRORS



TECHNICAL HIGHLIGHT

BOOKING LOGIC: CHECK AVAILABILITY

CREATE BOOKING

THROW EXCEPTION IF UNAVAILABLE

WHY CHOSEN : PREVENTS DOUBLE BOOKING

ENSURES SYSTEM RELIABILITY

LIVE DEMO

1. ADD / VIEW ROOM

2. CREATE BOOKING

3. SHOW BOOKING

4. ERROR CASE

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PS C:\Users\Priyam Dey\Downloads\Java Folder\term-ii-project-submission-Deypriyam> java -cp out hotel.ui.HotelApp
╔═══════════════════════════╗
║      HOTEL BOOKING MANAGEMENT SYSTEM   ╖
          Spring 2026 Project            ╖
╚═══════════════════════════╝

======= MAIN MENU =======
 1. View All Rooms
 2. View Available Rooms
 3. Register Guest
 4. Book a Room
 5. Cancel Booking
 6. Check-Out
 7. View Booking Details
 8. View Guest Booking History
 9. Admin Panel
 0. Exit
=========================
Enter choice: 1

===== ALL ROOMS =====
Room #101 [SINGLE] - 1 Bed, 1 Guest Max | Price: Rs.1500.0/night | AVAILABLE
Room #102 [SINGLE] - 1 Bed, 1 Guest Max | Price: Rs.1500.0/night | AVAILABLE
Room #103 [SINGLE] - 1 Bed, 1 Guest Max | Price: Rs.1500.0/night | AVAILABLE
Room #201 [DOUBLE] - 2 Beds, 2 Guests Max | Price: Rs.2500.0/night | AVAILABLE
Room #202 [DOUBLE] - 2 Beds, 2 Guests Max | Price: Rs.2500.0/night | AVAILABLE
Room #203 [DOUBLE] - 2 Beds, 2 Guests Max | Price: Rs.2500.0/night | AVAILABLE
Room #301 [SUITE] - King Bed, Living Area, Up to 4 Guests | Price: Rs.5000.0/night | AVAILABLE
Room #302 [SUITE] - King Bed, Living Area, Up to 4 Guests | Price: Rs.5000.0/night | AVAILABLE

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 8. View Guest Booking History
 9. Admin Panel
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=========================
Enter choice:
```

THANK YOU

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